

Mateusz Płoszaj-Mazurek

Computational designer · Grasshopper, simulation and BIM tooling · Architect, PhD

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(<https://github.com/Curiosit>)

Architect who has taught Grasshopper for seven years and builds the tools around it: plugins, simulation workflows, IFC pipelines and web viewers, for employers and clients in Denmark, Liechtenstein, Finland and Poland. I take a repetitive design task, automate it end to end and hand the team something they can run without me.

Commercial and tool work

2025	Free Drawing, Carbon Designer 3D One Click LCA Led a feature letting architects sketch building massing directly in the browser, from user problem through specification to release.
2024	IFC viewer for historical building objects client, Liechtenstein Browser viewer built with Three.js, React and TypeScript for exploring IFC models of historical building elements.
2025–26	OmegaFlux Per-point daylight prediction from geometric view descriptors with gradient-boosted trees, exported to ONNX and run in the browser. Dataset generation, held-out validation against Radiance, transfer to unseen room shapes. curiosit.github.io/OmegaFlux (https://curiosit.github.io/OmegaFlux/)
before 2024	Freelance AEC development clients via Upwork Short commercial jobs including IFC-to-web implementation.
2016–2024	In-house design tools Bjerg Arkitektur Grasshopper and scripting tools used in live projects, built end to end; daylight simulation studies for VELUX-related projects.
2023–24	goeko.gha: Ökobaudat in Grasshopper open source Grasshopper plugin that reads environmental product declarations from the German Ökobaudat database through its API. MIT licence, published on Food4Rhino. github.com/Curiosit/goeko.gha (https://github.com/Curiosit/goeko.gha)
2024	Wall carbon footprint configurator ShapeDiver No-code web app built from a Grasshopper definition, loading material data from Ökobaudat through an API.
2023–24	AEC Hackathons Copenhagen 2023 (SpeckleLCA), Zurich 2024 (InfraWeb), Copenhagen 2024 (OpenDetail, Best Mashup Project).

Experience

current	Researcher and lecturer Faculty of Architecture, Warsaw University of Technology - Daylight regulation, simulation and surrogate-model research; see publications. - Teaches Grasshopper and multi-criteria building optimization; coordinates a lecture cycle on public buildings; runs a second-year design studio.
2024–2026	Software engineer, interim product owner One Click LCA, Helsinki · remote - Built features for Carbon Designer 3D, an early-design tool for architects (TypeScript, React, Three.js, AG Grid). - Interim product owner in the second half of 2025; took the Free Drawing feature from specification to release. - Wrote methodology specifications and brought in a paying customer.
2024	Contract developer enxess, Liechtenstein Web viewer for IFC models of historical building objects (Three.js, React, TypeScript).

	2023–2024	Founder SLAD.AI Self-funded venture building early-design carbon estimation software; designed, built and tested with users.
	2014–2024	Architect, Certified Passive House Designer Bjerg Arkitektur · Hjørring 2014–16, Warsaw 2016–24 - Single- and multi-family housing, schools, kindergartens and offices, from 100 m² to 15,000 m². - All design phases: concept, permit and construction documentation, and architectural supervision on site. - Code compliance in practice under Danish and Polish regulations and Passive House certification; daylight simulation studies with VELUX. 2022–2024: architecture, sustainability and software consultant, building in-house design tools; co-author of the Polish whole-life carbon roadmap.
Skills	Design tools	Rhino and Grasshopper, Ladybug and Honeybee, Radiance, THERM, IFC and BIM, That Open Company tools
	Methods	Evolutionary and multi-objective optimization, surrogate modelling, gradient-boosted trees, dataset and validation design, parametric experiments
	Building domain	Climate-based daylight (sDA, ASE, UDI, EN 17037), insolation, Polish and Danish building regulations, Passive House, building carbon footprint
	Code	Python (NumPy, pandas, scikit-learn, XGBoost, ONNX), TypeScript, React, Three.js, Git, CI/CD
	Languages	Polish (native), English (fluent)
Teaching	7 years	Introduction to Grasshopper Parametric modelling and multi-criteria optimization of energy, daylight and carbon.
Grants and awards	2024	First prize, Best Mashup Project · AEC Hackathon, Copenhagen
	2023	Rector's team award for teaching achievements (third degree) · Warsaw University of Technology, academic year 2021/22
	2022	PLGBC Green Building Awards: best green building without certification · Passive House kindergarten in Podkowa Leśna, Bjerg Arkitektur; award received on behalf of the practice. Also a distinction for an educational building.
Education	2022	PhD in Architecture Warsaw University of Technology
	2014–2016	MSc in Architecture Aalborg University, Denmark
	2023–2024	Certifications Master BIM Software Developer (That Open Company, 2024) · Harvard CS50x (2023) · Certified Passive House Designer
Talks and panels	2023	Pro-environmental design: training sessions for architects · Saint-Gobain · two sessions, Jura Krakowsko-Częstochowska and Żnin, October 2023
Publications	A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland with A. M. Tofiluk · Journal of Building Engineering 120, 115397 (2026) · doi · open access	